Final Project Submission

DeepCompile: AI-Driven Compiler Optimization

Roll Number: 24CSB0A06 · Date: April 19, 2026

Phase 3 Completion Report

Covered Weeks

Weeks 9-12 under the revised project scope:

- source-to-source optimization only
- deterministic rule-based evaluation as the formal evidence path

Implemented Outcomes

Week 9

- Added `RefinementAgent`
- Added `OrchestratorAgent`
- Implemented bounded retry and escalation metadata
- Implemented workflow-level statuses and stop reasons

Week 10

- Switched the canonical CLI path toward orchestration by default
- Added shared explainability contract
- Added Markdown and HTML exporters

Week 11

- Added domain-specific rule modules for:

- scientific and numeric loops

- memory and buffer-heavy code

- Added in-memory caching for repeated analyses
- Added stage timing metadata to agent outputs

Week 12

- Defined a formal failure taxonomy document
- Added deterministic orchestration/reporting/domain tests

Validation Completed in This Revision

- `python -m py_compile` passed for the updated backend and CLI files
- `python -m unittest tests.test_orchestration_reporting -v` passed

Remaining Work for Weeks 13-14

- expand deterministic evaluation artifacts
- update user/developer manuals and submission summaries
- assemble final presentation and combined dossier